

STUDENT BOOK ANSWERS

Chapter 3 Metallic, ionic and covalent structure bonding

Worked example 3.1

- a Carbon has 4 valence electrons and a valency of 4. Chlorine has 7 valence electrons and has a valency of 1. The ratio would be 1C : 4Cl
- b Hydrogen has only one electron and needs 1 more electron (to fill the $1s^2$ orbital). Boron has three valence electrons and has a valency of 3. The ratio would be 1B : 3H.
- c Oxygen has 6 valence electrons and needs 2 more to have a filled outer shell so it has a valency of 2. Magnesium has two valence electrons. It needs to lose these 2 electrons to get a full outer shell. The ratio would be 1Mg : 1O.
- d Aluminium has a valency of 3. It needs to lose three to get a complete outer shell. Sulfur has 6 valence electrons and needs 2 more to have a filled outer shell. The lowest common multiple is 3×2 . So the ratio is 2Al : 3S.

Question set 3.1

- 1 Positive and negative charges in atoms provide the electrostatic forces of attraction or the electrostatic force that hold the atoms together.
- 2 Metallic, ionic and covalent.
- 3 The factors are the position on the periodic table and what element it combines with. Metals will form metallic bonds, non metals will form covalent bonds. When a metal and non metal are present; an ionic bond is formed.
- 4 The position in the periodic table indicates the number of valence electrons present. Elements in group 1 and 2 have 1 and 2 valence electrons respectively. Group 3 to 12 have a variable number of valence electrons and valency. Group 13 to 18 elements have 3 to 8 valence electrons respectively.

For elements with less than 4 valence electrons the valency is the number of valence electrons. When the number of valence electrons is more than 4 the valency is $(8 - \text{number of valence electrons})$. Oxygen's number of valence electrons is 6 and the valency is $8 - 6$, or 2.

- 5 It indicates the number of bonds the atom can form.
- 6
 - a Helium: 0
 - b Sodium: 1
 - c Phosphorous: 5
 - d Boron: 3
 - e Calcium: 2

- 7 a Silicon and hydrogen: 1:4
b Sodium and nitrogen: 3:1
c Aluminium and fluorine: 1:3
d Phosphorous and oxygen: 2:3

Question set 3.2

- 1 The valency electrons are loosely held, as shown by their low ionisation energy. These valence electrons are attracted to the nuclei of the other atoms and detach from their atoms. The electrons are delocalised.
- 2 Metallic bonding is the electrostatic attraction of delocalised electrons to the cations in the metallic lattice structure.
- 3 The ionisation energy is low for metal valence electrons. The valence electrons are loosely held. This means that they can be attracted to the nuclei of other nuclei. They can continue to be attracted to neighbouring nuclei in a random direction so they are delocalised.
- 4 a Metals can be drawn into a wire due to the delocalised electrons. When metal is drawn into wires: the cations slide over each other and the delocalised electrons between the two layers continue to be attracted to the nuclei. This provides an attractive force between the two layers which allow the metal to be drawn into a wire.
- b A metal fork is able to conduct electricity. The delocalised electrons will move through the fork as they are alternately attracted to then repelled when the fork is placed into the active pin of an AC power socket. This means a current will run through the fork.
- c A metal spoon in a cup of hot water can become hot enough to burn. The delocalised electrons rapidly transfer heat energy from the hot spoon to your hand when it is picked up.
- 5 a Good conductor of electricity; ductile
b Hard; high tensile strength
c Good conductor of heat; malleable
d Malleable; lustrous
e Good conductor of electricity; very high melting temperature
- 6 a The thermal conductivity and electrical conductivity both increase as you go down the table. The increase in thermal conductivity is more dramatic than the electrical conductivity. Both show the relationship that: As the size of the atom increases the thermal and electrical conductivity increases.
- b The first three elements are in the same period of the periodic table. As the number of valence electrons increases from 1 to 3 for sodium to aluminium respectively there are more delocalised electrons available so that the thermal conductivity and electrical conductivity can increase. More electrons can acquire the heat to transfer it to the other area and more electrons are free to move. Copper is a transition metal with filled 3d orbitals and a 4s¹ electron. The 4s¹ electron is further from the nucleus and is less bound. This increases the electrical and thermal conductivity of copper.
- c Melting point indicates the strength of the bonding between the atoms. As you move across the period the melting point increases as shown by the increase from sodium to aluminium. Transition metals have higher melting points compared to other metals. Transition metals have higher density than the other metals. This increases copper's melting point as the bonds between the atoms are closer and stronger.

Experiment 3.1: Comparing the properties of a compound with those of its component elements

Results

Properties	Magnesium	Oxygen	Magnesium oxide
Physical state	<i>Solid</i>	<i>Gas</i>	<i>Solid</i>
Colour	<i>Silver</i>	<i>Colourless</i>	<i>White solid</i>
Electrical conductivity	<i>Yes</i>	<i>No</i>	<i>No</i>
Solubility in water	<i>No</i>	0.0359 g O ₂ per L water at 25°C	<i>No (slightly)</i>
Reaction with HCl	<i>Yes</i>	<i>No</i>	<i>Yes</i>
Melting point	650°C	−219°C	2 852°C
Density	1.738 g cm ^{−3}	1.42 g /L	3.58 g cm ^{−3}

Analysis of results

The properties of the elements are very different to that of the compound. Magnesium is a metal, oxygen is a gas and magnesium oxide is an ionic solid. Magnesium and magnesium oxide are not soluble in water. Oxygen is slightly soluble in water. The magnesium oxide has a very high melting point.

Discussion

Other properties like heat of vaporisation or the chemical resistance or chemical activity could also be compared. However it is likely that these would also show differences between the elements and the compounds.

Conclusion

Ionic compounds have properties different to the elements that make up the compounds.

Question set 3.3

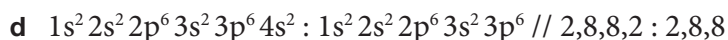
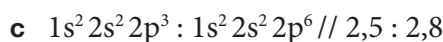
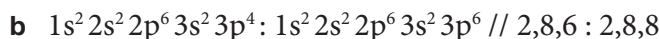
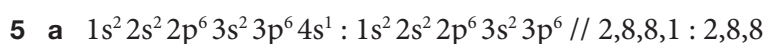
- 1 Ionic bonding is due to the electrostatic attractions between oppositely charged ions.
- 2 Metals have a low affinity for electrons and low ionisation energy. They tend to lose the valence electrons. Non metals have a high affinity for electrons, high ionisation energy and are more electronegative than metals. Non metals tend to gain electrons. This is why metals donate electrons to become cations. And why non metals accept electrons to become anions.
- 3 High melting point. The electrostatic attraction between the anion and cation is strong so a large amount of energy is required to melt the ionic substance.

Conducts electricity in the molten state and in solution: The solid state consists of a lattice of alternating cations and anions. When in the molten state or when in solution these charged particles are free to move. This allows electricity to be conducted.

Brittle: the ions are held rigidly in place in the lattice. When under stress the ions of similar charge line up. As these ions repel each other the crystal can shatter.

Mostly soluble in water. Water molecules can attract the charged ions. If this occurs the salt can dissolve.

- 4 Ionic lattices are gigantic lattices of repeating ions. The electrostatic attraction between the positive and negative charge means that ions of opposite charge surround each other in a fixed pattern; this ratio represented by the empirical formula.



6 Nitrogen atom:



Nitrogen ion:



Calcium atom:



Calcium ion:



7 a +2

b -3

8 a Sodium ion has a charge of +1, bromide ion has a charge of -1. The ionic formula has to be neutral so the ratio has to be 1 : 1.

b Aluminium ion has a charge of +3. For the salt to be neutral the anion needs to provide 3 charges. Fluorine ion has a charge of -1. Three fluorine ions are needed to make the salt neutral. Although Al_2F_6 has the correct ratio, AlF_3 is the empirical formula.

9 a Mg^{2+} and O^{2-}

b Ag^+ and Cl^-

c Li^+ and I^-

10 Elements are more stable when the outer shell is filled or empty. Hydrogen can do this by either losing one electron or gaining one electron.

Worked example 3.3

a Ions present: $\text{Na}^+ \text{F}^-$

Compare charges: both one

NaF

b Ions present: $\text{Mg}^{2+} \text{O}^{2-}$

Compare charges: both two

MgO

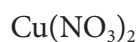
c Ions present: $\text{K}^+ \text{OH}^-$

Compare charges: both one

KOH

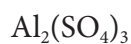
d Ions present: Cu^{2+} NO_3^-

Compare charges: one 2+ and 1–



e Ions present: Al^{3+} SO_4^{2-}

Compare charges: lowest common multiple of 3 and 2 is 6 ($+3 \times 2$ and -2×3)



Question set 3.4

1 a Ionic compounds have no overall (net) charge.

b The number of charges on the cations will cancel out the number of charges on the anions.

2 a Bromide: Br^-

b Sulfide: S^{2-}

c Barium: Ba^{2+}

d Potassium: K^+

e Nitride: N^{3-}

3 a Bromide:



c Barium:



e Nitride:



4 a KI: Potassium iodide

b BaCl_2 : Barium chloride

c CaH_2 : Calcium hydride

d PbO_2 : Lead (IV) oxide

e Na_3PO_4 : Sodium phosphate

f $\text{Zn}(\text{NO}_3)_2$: Zinc nitrate

5 a Magnesium nitride: Mg_3N_2

b Aluminium oxide: Al_2O_3

c Mercury(II) sulphide: $\text{Hg}(\text{II})\text{S}$

d Ammonium hydroxide: NH_4OH

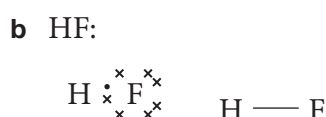
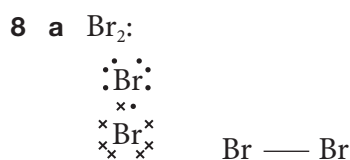
e Copper(I) carbonate: Cu_2CO_3

f Zinc phosphate: $\text{Zn}_3(\text{PO}_4)_2$

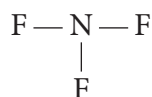
- 6 a +1
b +2
c +1
d +1
e -1
f -1
g -2

Question set 3.5

- 1 a A covalent bond is an electrostatic attraction between shared electrons and nuclei of non-metal atoms.
- b i Two
ii Four
iii Six
- 2 Carbon and silicon
- 3 a 8, 2 and 8 respectively
b Ne, He and Ar
- 4 Covalent network substances have the strong covalent bond within the whole network. This explains their very high melting points. The strong bonds mean that the substances are extremely hard and brittle. They are insoluble in water and most other solvents, as the bonds are strong. Covalent molecular substances have weak intermolecular forces between the molecules. These weak bonds do not take much energy to break so the molecules have a low melting point. As these forces are weak the molecules can be easily moved relative to each other making them malleable and soft rather than brittle. The ability of the molecules to be soluble will depend upon the type of intermolecular bonds that are present.
- 5 Graphite can conduct electricity as there are delocalised electrons which can move along the layer. All the electrons in diamond are involved in covalent bonding and are not free to move. This explains why diamond cannot conduct electricity.
- 6 a F_2O
b BH_3
- 7 Carbon has four valence electrons and requires four more electrons to complete the outer shell. Hydrogen has only one electron that can be shared. So four hydrogens will combine with carbon rather than 3 or 5, forming CH_4 .



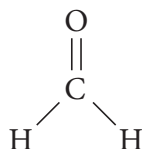
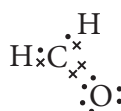
c NF_3 :



d CS_2 :



e HCHO



- 9 There are 4 bonding pairs of electrons between the Si and Cl atoms. There are also 3 pairs of non-bonding electron pairs around each Cl.
- 10 For an odour to be detected the substances have to be airborne. Covalent molecular substances have weaker intermolecular bonds and are much more likely to be gases than other substances.

Question set 3.6

- 1 The first named element is the one that is further to the left in the periodic table. If both elements are in the same group then the largest one (lower down the group) is named first. The exception is when oxygen is bonded to the halogens (Cl, Br, I) as oxygen is named last.
- 2 a Sulfur dioxide: SO_2
b Dinitrogen pentoxide: N_2O_5
c Carbon tetrachloride: CCl_4
d Nitrogen trifluoride: NF_3
e Silicon tetrabromide: SiBr_4
- 3 a N_2O : Dinitrogen oxide
b NCl_3 : Nitrogen trichloride
c SO_3 : Sulfur trioxide
d H_2S : Dihydrogen sulfide
e N_2O_4 : Dinitrogen tetraoxide

Experiment 3.2: Comparing different types of substances

Conclusion

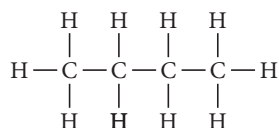
Metals conduct electricity. Salt and strong acid solutions can conduct. Molten salts can also conduct electricity.

Question set 3.7

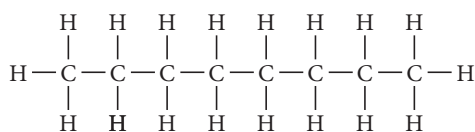
- 1 a Alkanes: $\text{C}_n\text{H}_{2n+2}$
b Alkenes: C_nH_{2n}
c Benzene: C_6H_6

- 2 The difference between saturated and unsaturated compounds is the type of bonds that may be present. A saturated compound has only single bonds. An unsaturated compound will contain one or more double bonds.
- 3 The cyclic delocalisation of electrons in the benzene molecule makes it extremely stable. When benzene undergoes reactions the stable ring is preserved.
- 4 C_6H_{12} is an alkene. The general formula C_nH_{2n} is observed ($6 \times 2 = 12$).
 C_6H_{14} is an alkane. The general formula C_nH_{2n+2} is observed.

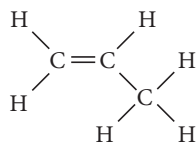
5 a Butane



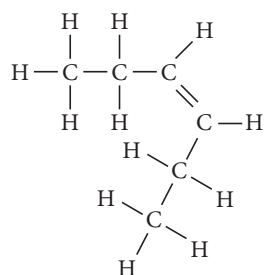
b Octane



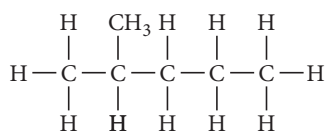
c 1-Propene



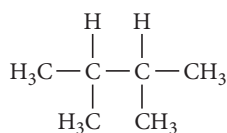
d 3-Hexene



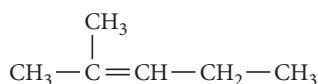
e 2-Methylpentane



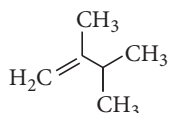
f 2,3-Dimethylbutane



g 2-Methyl-2-pentene



h 3,3-Dimethyl-1-butene



- 6 a** Alkanes and alkenes are both covalent molecules. The bonds between the molecules are weak so they have similar melting points and boiling points. As they have only carbon and hydrogen in the molecule they have similar solubility in water; i.e they are insoluble in water but will dissolve in other organic solvents.

The chemical reactivity is different as alkenes have at least one double bond. The double bond is very reactive and allows the alkenes to undergo addition reactions. Alkanes without the double bond are not able to undergo addition reactions.

- b** Benzene is essentially insoluble in water. Water molecules are much more strongly attracted to other water molecules than they are to benzene. Like the alkanes and alkenes, benzene only has carbon and hydrogen in the molecule, so it too is insoluble.

7 a Butane

b Nonane

c 2-pentene or pent-2-ene

d 1-butene or but-1-ene

e 3-methylhexane

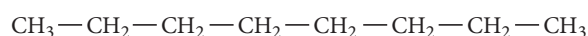
f 4-ethyl-3methylheptane

g 3-methyl-2-pentene or 3-methyl pent-2-ene

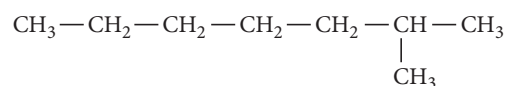
h 2,3-dimethyl pent-1-ene

- 8** Addition reactions occur over a double bond. The other reactant is added to either side of the double bond. In substitution reactions one atom is replaced or substituted for another. The substituted atoms form a separate product.

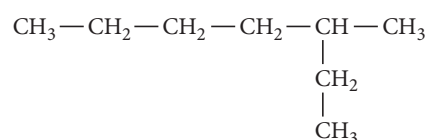
9 Octane



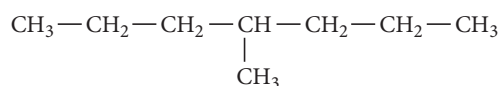
2-Methylheptane



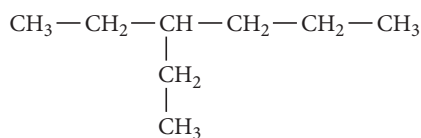
3-Methylheptane



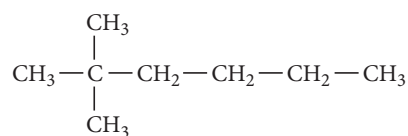
4-Methylheptane



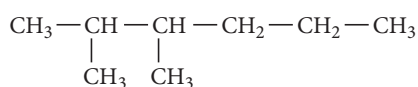
3-Ethylhexane



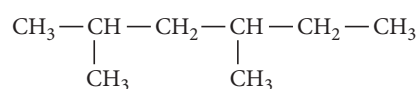
2,2-Dimethylhexane



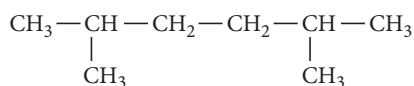
2,3-Dimethylhexane



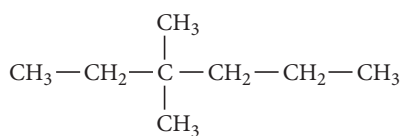
2,4-Dimethylhexane



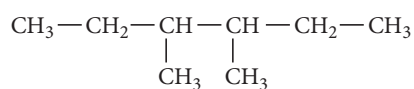
2,5-Dimethylhexane



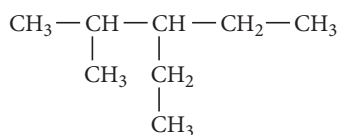
3,3-Dimethylhexane



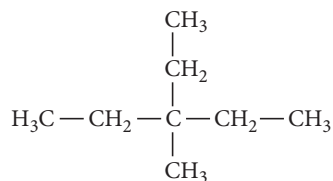
3,4-Dimethylhexane



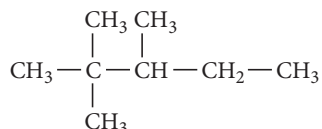
3-Ethyl-2-methylpentane



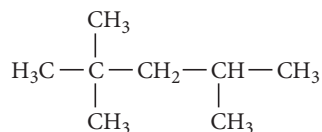
3-Ethyl-3-methylpentane



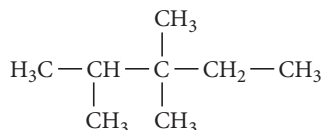
2,2,3-Trimethylpentane



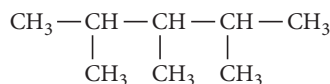
2,2,4-Trimethylpentane



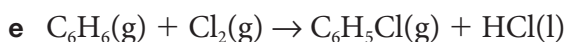
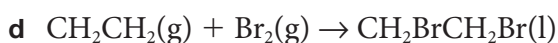
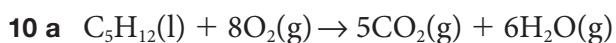
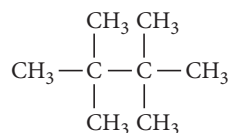
2,3,3-Trimethylpentane



2,3,4-Trimethylpentane



Tetramethylbutane



- 11** Bromine will react with an alkene and will add across the double bond. Bromine is a reddish brown coloured liquid. Once Bromine has reacted the colour must disappear. If the gaseous hydrocarbon is bubbled through bromine and the colour is reduced then it can be concluded that the hydrocarbon is an alkene or has unsaturated bonds.

Chapter review questions

1 a Covalent, ionic and metallic

- b** Metallic bonds occurs in metals. The electrostatic attraction between the delocalised electrons and the metal atoms that have released them (cations). See Figure C3.4 on p. 164 of the student book.

Ionic bonding: the electrostatic attraction between metal atoms that have lost electrons (cations) and non metal atoms that have gained electrons (anions). See Figure C3.15 on p. 174 of the student book.

Covalent bonding: the electrostatic force of attraction between electrons and the nucleus of more than one atom holds the atoms together. It occurs between non metal elements. See Figure C3.16 on p. 179 of the student book.

Properties/substances	Metallic	Ionic	Covalent molecular	Covalent network
Electrical conductivity	Yes	When solid - No Yes for molten state and solutions	No	No Graphite: yes
Melting points and boiling point	High – although there are exceptions such as Hg, main group metals have lower temps compared to the transition metals	High	Various many are liquids and gases at room temperature	High
Hardness	Most are fairly hard Gold, lead and calcium are soft	Hard and brittle	Softer	Extremely hard and brittle

- 3** Diamond has carbon atoms arranged in a tetrahedral network arrangement. Diamond is very hard. Each carbon covalently bonds to four other carbon atoms. Covalent bonds are strong so a lot of energy is required to break them; this accounts for the high melting and boiling point. Diamond is very hard as the network structure means that the forces are shared over the whole network. There are no free charged particles to conduct electricity.

Graphite is a covalent network in which the carbon atoms are arranged in flat parallel layers. Each carbon atom is covalently bonded to three other carbon atoms. Covalent bonds are strong so a lot of energy is required to break them; this accounts for the high melting and boiling point. The fourth electron is delocalised and is free to move. This allows graphite to conduct electricity. The flat layers have only weak bonds between the layers. The weak bonds are easy to break which makes graphite soft.

Fullerene molecules have carbon covalently bonded to three other carbons. The fourth electron is delocalised. The fullerene carbons are arranged in pentagonal and hexagonal rings. These rings make molecules non- planar and they resemble open cage structures. They are unable to conduct electricity as the electrons cannot move from one molecule to the other. The weak forces between the molecules mean that the melting and boiling points are low.

The three allotropes of carbon have no attraction to water molecules so are all insoluble.

- 4** Alkanes, alkenes and benzene are covalent molecules. They have weak intermolecular forces between the molecules; resulting in low melting and boiling points and do not conduct electricity. They are hydrocarbons that have no attraction to water molecules. They are insoluble in water. They can be burnt in oxygen to form carbon dioxide and water.

The differences observed between these groups is due to the bonding of the molecules. There is a double bond in alkenes, a resonance structure of a flat ring in benzene and only single bonds in alkanes.

The presence of a double bond means that the alkenes are much more reactive than alkanes and benzene. Benzene is much more stable due to the resonance structure of the delocalised electrons in the ring.

Alkanes and Benzene undergo substitution reactions whilst alkenes undergo substitution and addition reactions.

- 5 Isomers are molecules with the same molecular formula but different structural formula, for example butane and methyl propane have different structures but have the same molecular formula C_4H_{10} .
- 6 Phosphorus trichloride has a low boiling point as it is a covalent molecule with weak intermolecular forces. There are no free particles to conduct electricity.

Calcium chloride is an ionic solid with a high melting point. This is due to the strong ionic bonds that require a large amount of energy to break. When this solid is melted the charged ions are free to move. The melted salt can conduct electricity.

- 7 Magnesium: metallic

Tetrabromomethane: covalent molecular

Barium chloride: ionic

Phosphorus triiodide: ionic

Silicon dioxide: covalent network

Lithium sulphide: ionic

Iodine: covalent molecular

Diamond: covalent network

- 8
 - a Silver oxide
 - b Aluminium chloride
 - c Potassium sulfate
 - d Ammonium bromide
 - e Magnesium hydroxide
 - g Iron (II) carbonate

- 9
 - a $MgCl_2$
 - b $CaCO_3$
 - c $Cu(NO_3)_2$
 - d NH_4Cl
 - e K_2S
 - f PbO_2

- 10
 - a
 - i +3
 - ii +2
 - b
 - i -1
 - ii -3

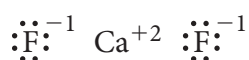
11 a Phosphorus trichloride

- b** Sulfur tetrafluoride
- c** Dinitrogen trioxide
- d** Nitrogen oxide
- e** Dichlorine heptoxide

12 a BCl_3

- b** PI_5
- c** HBr
- d** N_2O_4
- e** SiO_2

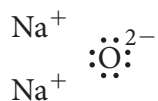
13 a CaF_2



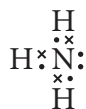
b HCl



c Na_2O



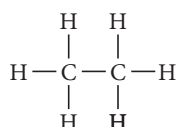
d NH_3

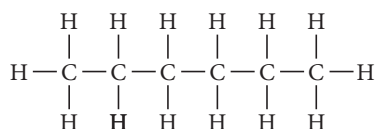


14 a Butane

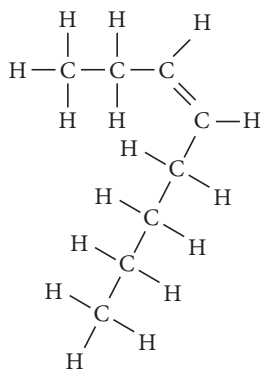
- b** Hexane
- c** 1 Pentene
- d** 2 butene or but-2-ene
- e** 2-pentene or pent-2-ene
- f** 2,3-dimethyl butane
- g** 3-ethyl hex-2-ene
- h** 2,3-dimethyl pent-2-ene or 2,3-dimethyl-2-pentene

15 a Ethane

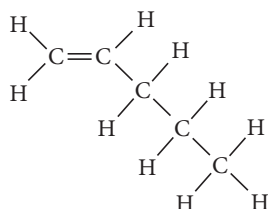


b Hexane

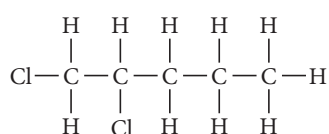
c 3-octene



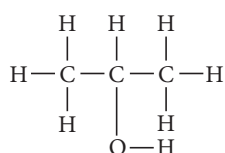
d 1-pentene



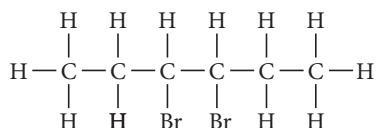
e 3,4-dimethyl-2-pentene



f 3-methyl-1-pentene



g 2-ethyl-3-methyl-1-hexene

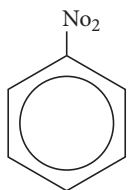
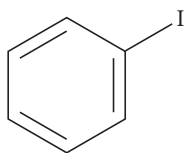
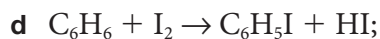
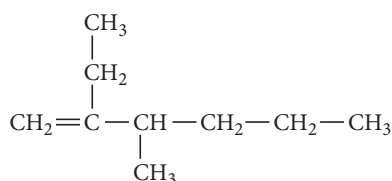
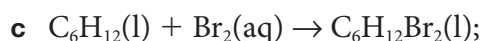
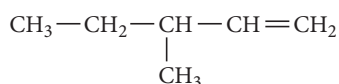
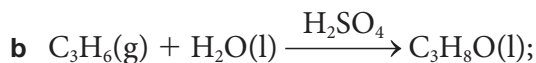
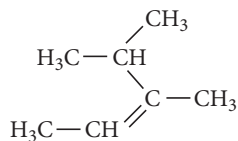
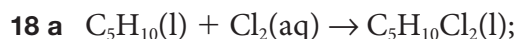
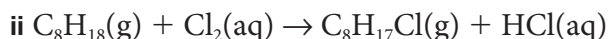
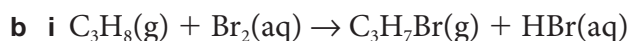


16 a 1-hexene. Numbering should start at the end of the molecule closest to the double bond.

b 3-heptene. Numbering should start at the end of the molecule closest to the double bond.

17 a i $\text{C}_5\text{H}_{12}(\text{g}) + 8\text{O}_2(\text{g}) \rightarrow 5\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$

$$\text{ii } 2\text{C}_7\text{H}_{14}(\text{l}) + 21\text{O}_2(\text{g}) \rightarrow 14\text{CO}_2(\text{g}) + 14\text{H}_2\text{O}(\text{g})$$



19 a A Metallic, B Covalent network, C Covalent molecular, D Ionic, E Covalent network

- b** Copper has delocalised electrons so can conduct electricity very easily; in both solid and liquid states. Sodium chloride has ionic bonds between the charged ions. The lattice structure means that the ions are not free to move. However, in the liquid state there are free moving charged particles, which means that the liquid can conduct electricity.
- c** Covalent molecular compounds have weak intermolecular forces. These do not require a lot of energy to break; these substances have low melting and boiling points. They do not have any free moving particles; they cannot conduct electricity. They may have no attraction to water molecules so may not be soluble in water.

Covalent network compounds have high melting and boiling points as they have strong covalent bonding between the atoms. Covalent bonds require a lot of energy to break. They do not have any free moving particles so they cannot conduct electricity. They have no attraction to water molecules so are not soluble in water.

- 20** A: Water is soluble in water, non flammable and doesn't react with Bromine.
- B: As it decolorises bromine in the presence of UV light it must have an unsaturated bond so it has to be hexene. This is consistent with being insoluble in water and flammable.
- C: Pentane: is a hydrocarbon which is insoluble in water and flammable. It must be a saturated molecule as it does not react with Bromine.
- D: Methylated spirits is an organic molecule which can dissolve in water. It is flammable. It must be a saturated molecule as it does not react with Bromine. It is also the last choice.